

Sensor Placement Benchmark — California 2021, Uniform Coverage Kernel

Wildfire Drone Routing Project

2026-03-13

1. Introduction

This report presents sensor and drone placement results for the California 2021 wildfire dataset at three budget levels (\$20M, \$100M, \$500M), using a **uniform coverage kernel** in the ILP formulation.

Dataset: 931 valid fires from the 2021 USFS California wildfire records, filtered for known date/time and valid WFPI data coverage.

Risk map: Pyrologix Ignition Probability (static, trained on 2006–2020 data, no data leakage for 2021).

Script: `run_benchmark_california2021_yearly.py`

2. Coverage Kernel

2.1 Problem with the previous kernel

The original placement ILP used a kernel derived from a random walk diffusion process. This kernel had two issues:

1. **Normalization bug:** The kernel was divided by the origin value, inflating weights so they summed to roughly the number of reachable cells (~49) instead of the patrol budget (~7). This caused the ILP to assign unreasonably many drones per station (up to 58–76) to push corner cells to full coverage.
2. **Wrong model:** A random walk models independent, uncoordinated drones. In practice, the routing stage (TOP / MaxCoverage) coordinates drones to cover non-overlapping regions, eliminating the diminishing returns of independent walks.

2.2 Uniform kernel

The corrected kernel models coordinated drone patrol:

- **Reachable zone:** $B \times B$ cells, where $B = \max_battery_time = 7$ (L_∞ distance $\leq \lfloor B/2 \rfloor = 3$ from the station).
- **Weight per cell:** $w = 1/B = 1/7 \approx 0.143$ for all 49 reachable cells.

- **Interpretation:** Each drone visits $B = 7$ distinct cells per charge cycle. With coordinated routing, B drones saturate the full zone ($B \times w \times B^2/B^2 = 1$).
- **ILP constraint:** $\theta_i \leq \sum_j w \cdot nc_j$, $\theta_i \leq 1$.

Properties:

- Sum of kernel weights = $B^2/B = B = 7$ (total patrol budget of one drone).
- Drones needed to saturate one station's zone = $B = 7$.

2.3 Bug fixes applied

Two additional bugs were identified and fixed in the ILP formulation:

1. **Masked cells in objective (fixed):** The risk map contained non-zero values at masked (infeasible) cells — snow, urban, non-burnable areas. Since the objective summed $\text{risk}_i \cdot \theta_i$ over all $261 \times 161 = 42,021$ cells, 57.6% of the total risk came from infeasible cells. This distorted the objective and wasted budget on coverage of areas where no fires occur.

Fix: `static_map .*= mask` — zero out risk for masked cells before building the ILP objective.

2. **Pre-filtering too aggressive (fixed):** A hardcoded top-20% candidate threshold restricted station candidates to ~2,238 cells, whose 7×7 zones could only reach 4,279 feasible cells (38% of 11,188). At \$100M this forced 181 stations into an area needing only ~88, causing 49.1% coverage overlap waste and only 39.6% of feasible cells covered.

Fix: `candidate_percentile = 0.00` — keep all feasible cells as candidates (solve time remains under 20 seconds).

3. Simulation Parameters

Parameter	Value
Grid	California 2021 WFPI/Pyrologix, 1309×805 (~1 km/cell)
Operational grid	261×161 (5×5 coverage blocks)
Drone speed	600 m/min
Coverage radius	2900 m (~2.9 km)
Battery (max)	1 h = 7 operational substeps
Transmission range	50 km
Scenarios benchmarked	100 random fires (seed=42) from 931 valid
Placement strategy	<code>SensorPlacementMaxCoverageGaussianTimeMaskedBudget</code>
Solver	Gurobi (academic licence, 600 s time limit)
Candidate filtering	None (all 11,188 feasible cells)

4. Routing Strategies

Two routing strategies are benchmarked on each placement.

Strategy 1 — DroneRoutingTOPMasked (TOP) PSO-based Team Orienteering Problem solver.

Parameter	Value
Reevaluation step	5 substeps
Optimization horizon	10 substeps
Time limit per call	60 s
Burn-map handling	static Pyrologix (tiled internally)
Visited-cell suppression	<code>reset_time = 2 × max_battery_time = 14</code> substeps

Strategy 2 — DroneRoutingMaxCoverageGrowingMasked (MaxCov) Greedy max-coverage routing with growing visited-cell suppression.

Parameter	Value
Reevaluation step	5 substeps
Optimization horizon	10 substeps
Time limit per call	60 s
Burn-map handling	static Pyrologix (tiled internally)
Visited-cell suppression	growing (cells visited during current horizon suppressed)

Both strategies use `burnmap_type="static"`: the single-frame Pyrologix map (1, 261, 161) is tiled internally to the full horizon length. Because the map never changes between scenarios, all clusters share a single routing log per strategy (`log_key="pyrologix"`).

5. Budget Comparison

5.1 Allocation summary

	\$20M	\$100M	\$500M
Ground sensors	0	0	12
Charging stations	40	200	1183
Drones	280	1400	2016
Budget used	\$20.00M	\$100.00M	\$279.45M
MIP gap	0.0%	0.01%	0.01%
Solve time	2.3 s	19.6 s	12.4 s
Clusters	9	1	1

All three placements are **provably optimal** (MIP gap $\leq 0.01\%$). For budgets above \$300M, the placement objective now includes a small regularization term $-0.1 \times \text{budget_used}$, so the 500M run prefers lower-cost solutions when coverage is already saturated.

5.2 Drone allocation

	\$20M	\$100M	\$500M
Stations with 7 drones	40 (all)	200 (all)	11
Stations with 0 drones	0	0	0
Stations with 1–6 drones	0	0	1172

At \$20M and \$100M, the ILP assigns exactly 7 drones to every station (full zone saturation). At \$500M, the budget regularization spreads the solution across many stations with mostly small per-station drone counts instead of forcing full 7-drone saturation everywhere.

5.3 Coverage efficiency (100M, before vs. after fix)

Metric	Before (top-20% filter)	After (no filter)
Stations	181	200
Feasible cells covered	4,436 (39.6%)	9,541 (85.3%)
Overlap waste	49.1%	0.4%
Objective achieved	48.1% of max	89.9% of max
Min station spacing (L_∞)	1	5

The fix eliminates nearly all overlap waste and more than doubles the number of feasible cells covered.

5.4 Key observations

\$20M: The optimizer allocates the full budget to stations (40) and drones (280), with zero ground sensors. At this budget level, drone coverage is more cost-effective than ground sensors. The 9 clusters target the highest-risk areas.

\$100M: Again, no ground sensors — all budget goes to 200 stations with 1,400 drones. The stations space themselves at $L_\infty \geq 5$ apart, covering 85.3% of all feasible cells with near-zero overlap. All stations merge into one connected cluster.

\$500M: With the new budget regularization active above \$300M, the optimizer now stops at \$279.45M instead of spending the full budget. The selected layout uses only 12 ground sensors, together with 1,183 charging stations and 2,016 drones, indicating that near-maximal coverage can be maintained without the previous dense blanket of ground sensors.

6. Cluster Structure

6.1 Cluster counts

	\$20M	\$100M	\$500M
Singleton clusters	variable	0	0
Multi-station clusters	variable	1	1
Total	9	1	1

A fire scenario is **discoverable** if its ignition point (in operational space) falls within L_∞ distance ≤ 3 of at least one charging station (one-way reach on a single charge).

7. Placement Maps

7.1 \$20M Budget

Data scale (1 km/cell)

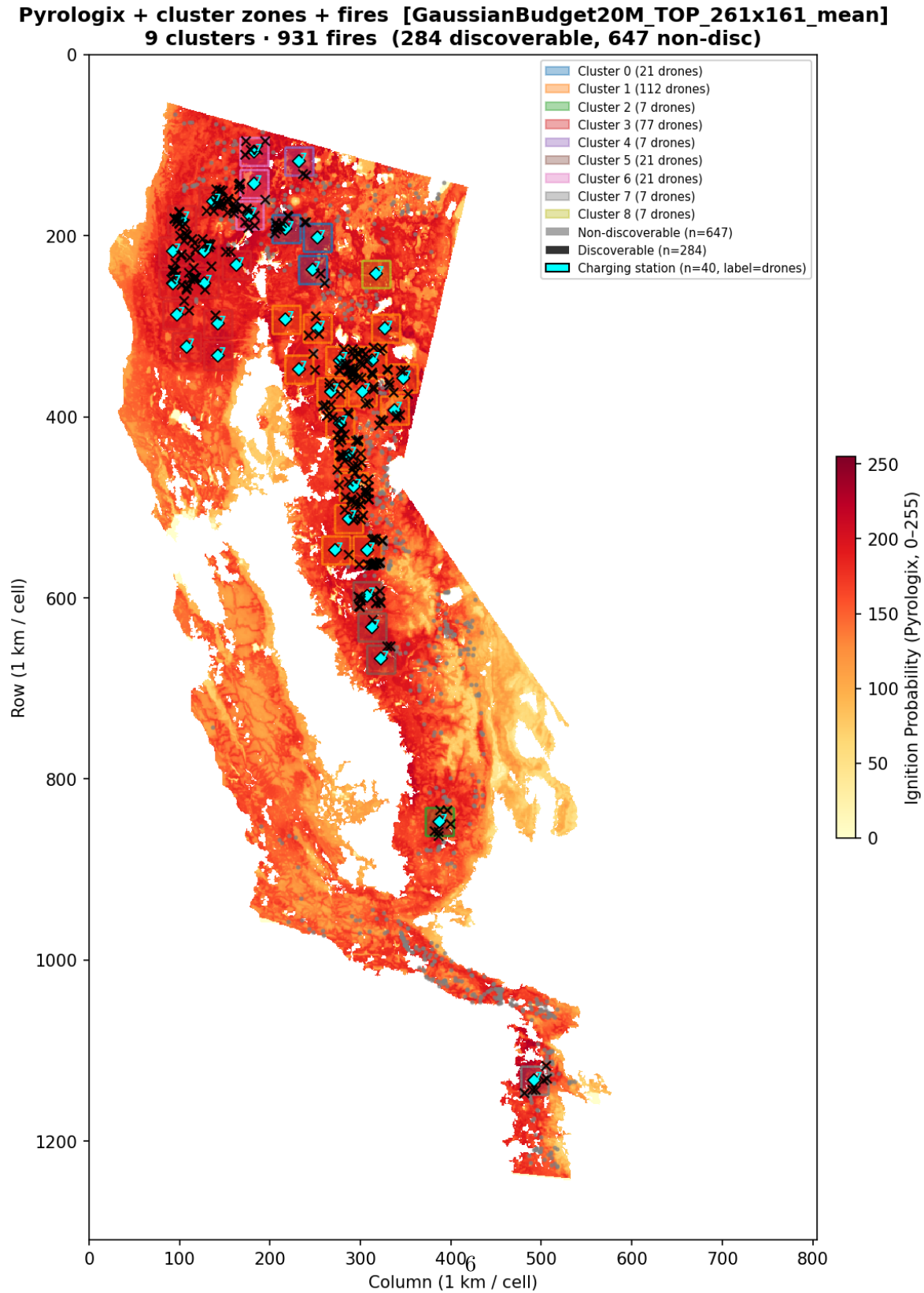


Figure 1: 20M placement — data scale. 9 clusters, 40 stations, 280 drones, 0 ground sensors. Cluster

Operational scale (5 km/cell)

**Pyrologix (opt scale) + cluster zones + fires [GaussianBudget20M_TOP_261x161_mean]
9 clusters · 931 fires (grid 261×161, 5 km / cell)**

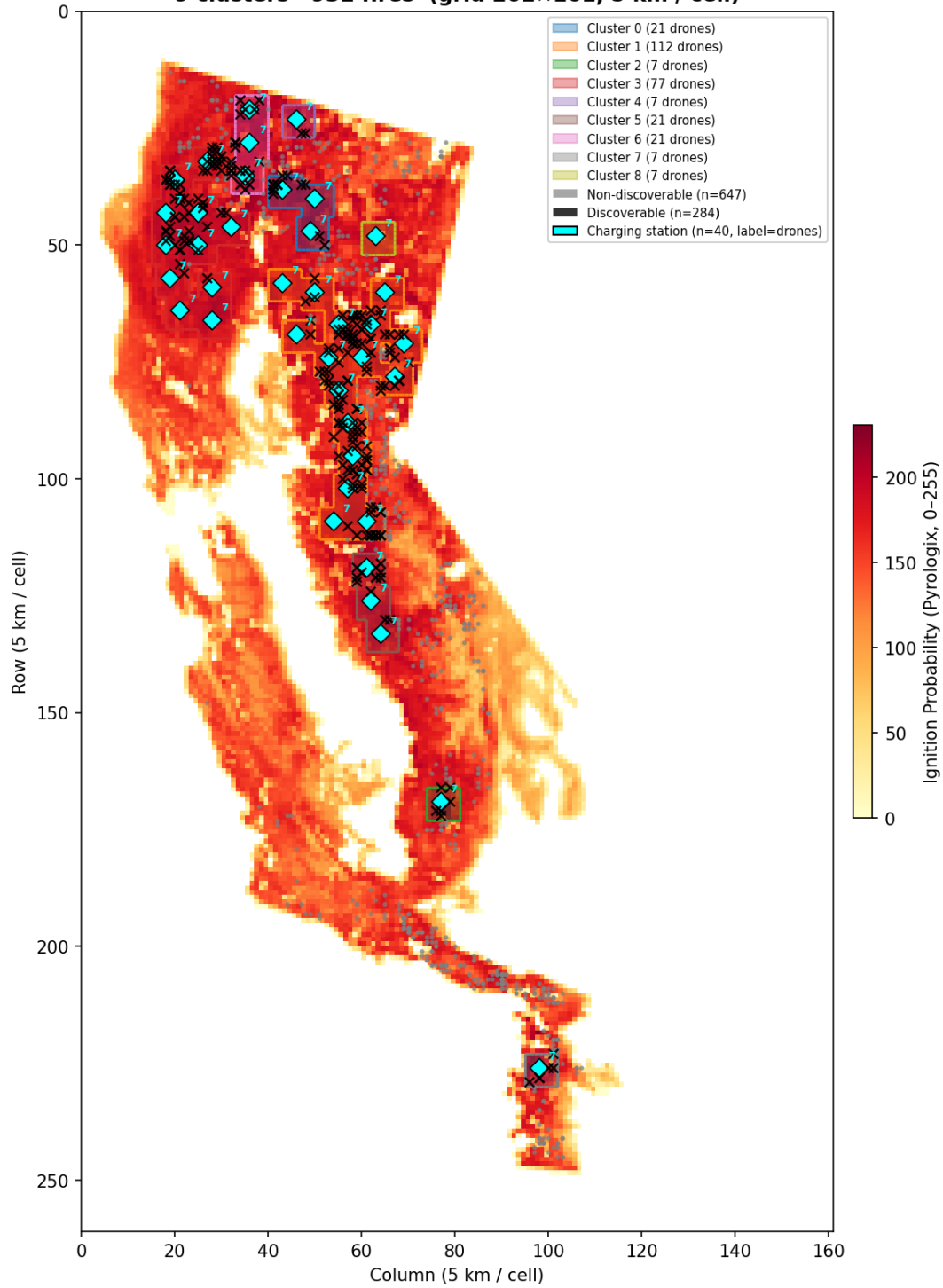


Figure 2: 20M placement — operational scale.

7.2 \$100M Budget

Data scale (1 km/cell)

Pyrologix + cluster zones + fires [GaussianBudget100M_TOP_261x161_mean]
1 clusters · 931 fires (918 discoverable, 13 non-disc)

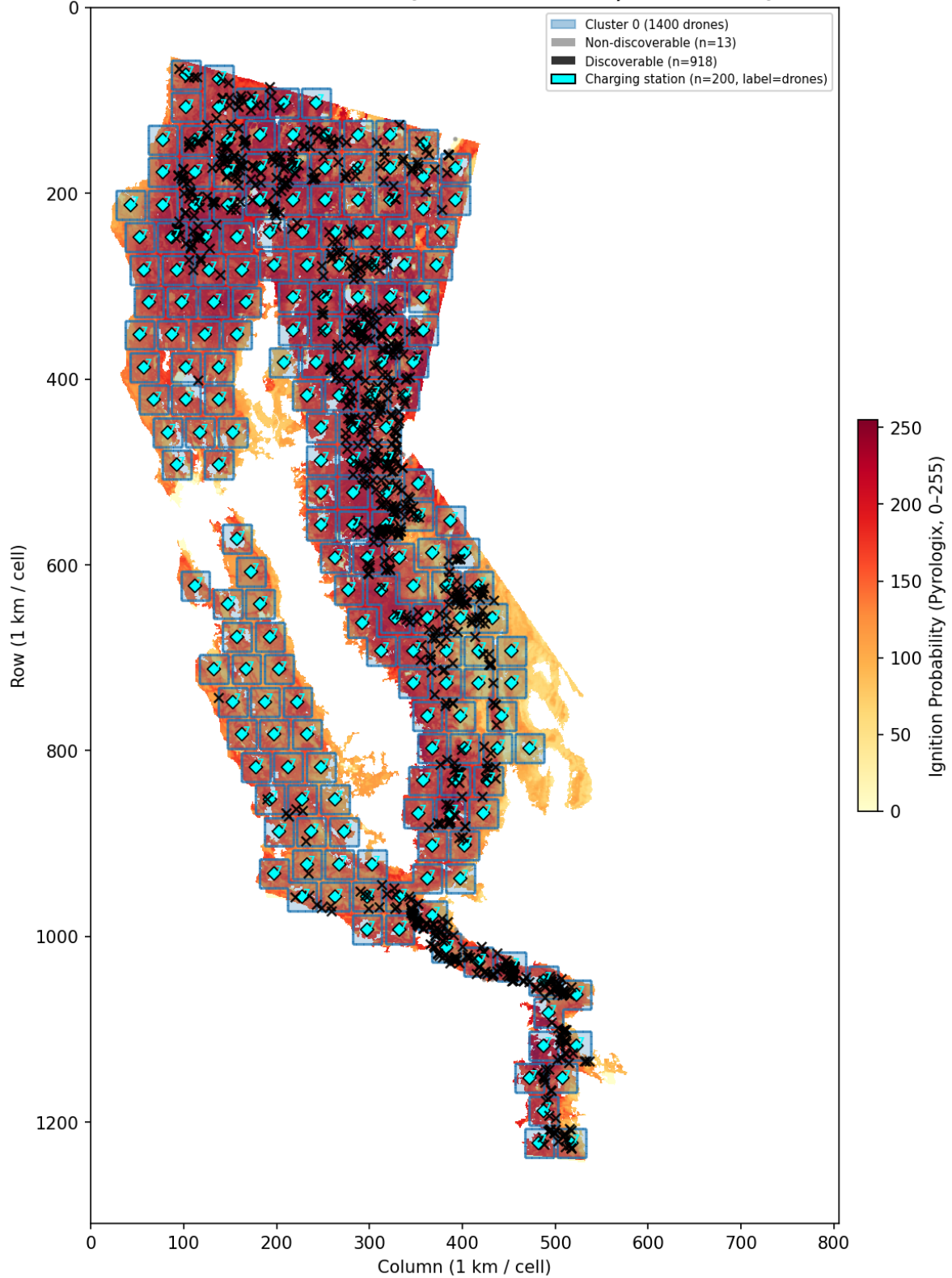


Figure 3: 100M placement — data scale. 1 cluster, 200 stations, 1400 drones, 0 ground sensors. Stations spaced at $L_\infty \geq 5$, covering 85% of feasible cells with near-zero overlap.

Operational scale (5 km/cell)

Pyrologix (opt scale) + cluster zones + fires [GaussianBudget100M_TOP_261x161_mean]
1 clusters · 931 fires (grid 261×161, 5 km / cell)

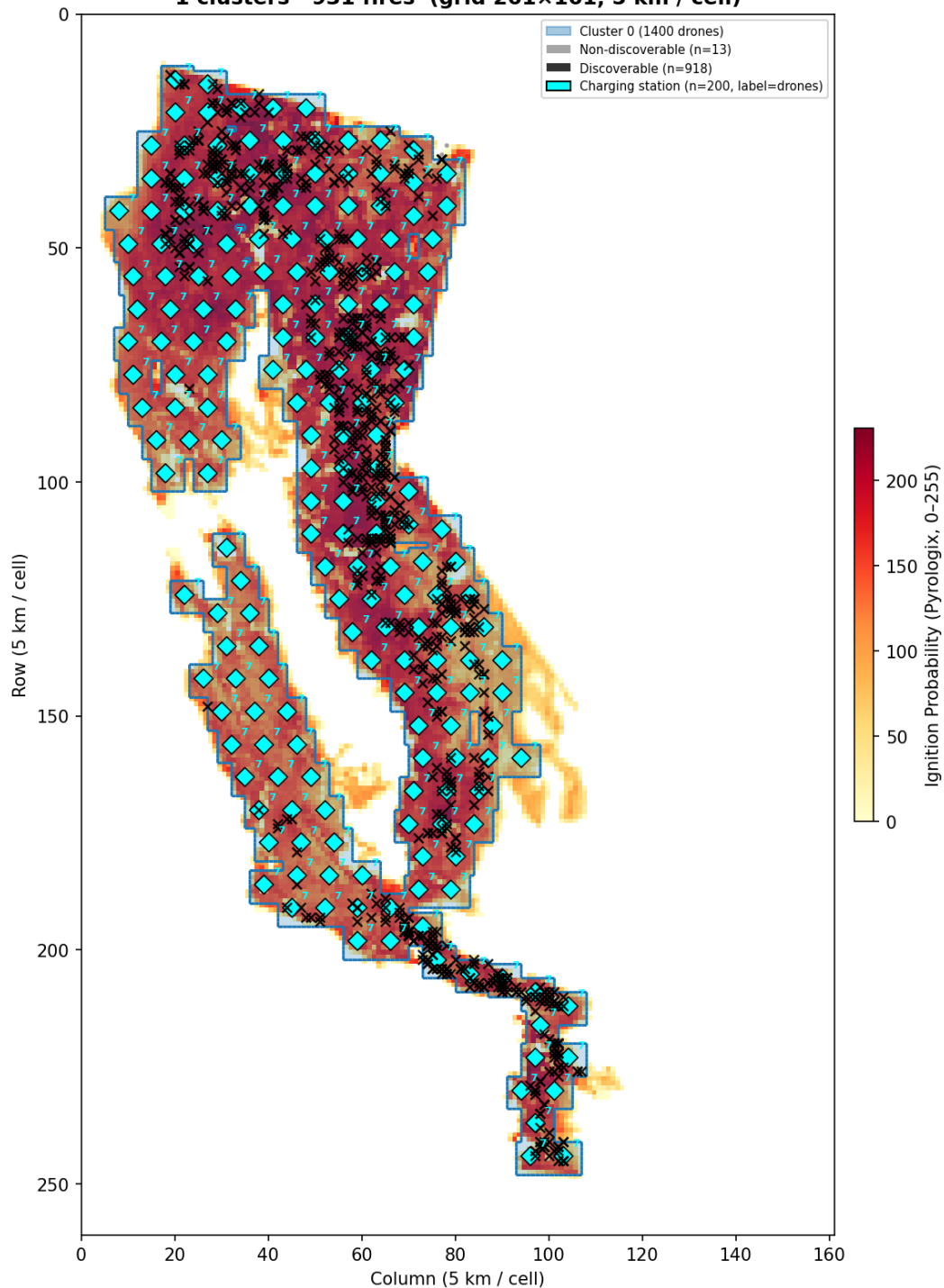


Figure 4: 100M placement — operational scale.

7.3 \$500M Budget

Data scale (1 km/cell)

**Pyrologix + cluster zones + fires [GaussianBudget500M_TOP_261x161_mean]
1 clusters · 931 fires (931 discoverable, 0 non-disc)**

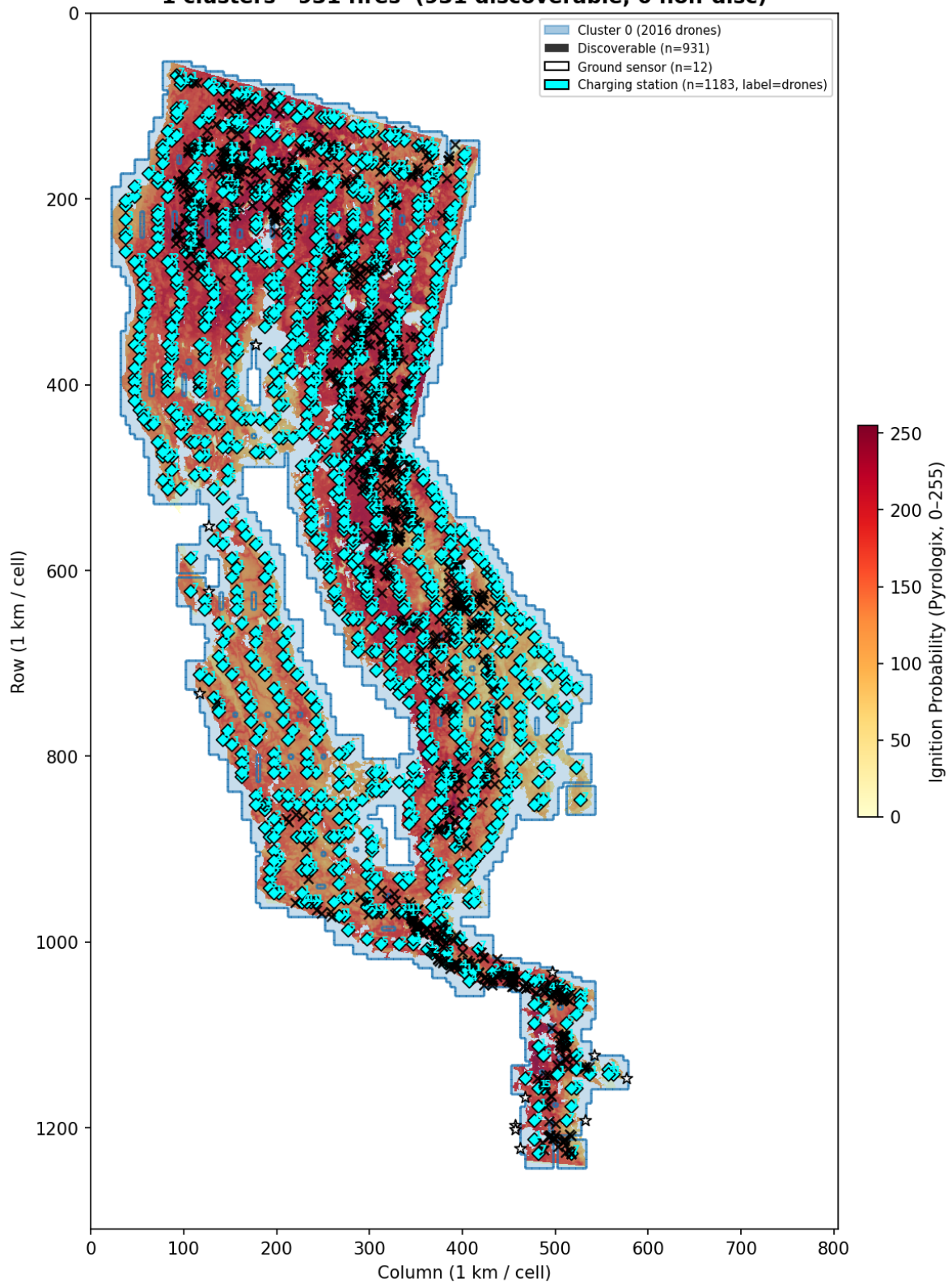


Figure 5: 500M placement — data scale. 1 cluster, 1183 stations, 2016 drones, 12 ground sensors. Budget used: \$279.45M.

Operational scale (5 km/cell)

Pyrologix (opt scale) + cluster zones + fires [GaussianBudget500M_TOP_261x161_mean]
1 clusters · 931 fires (grid 261×161, 5 km / cell)

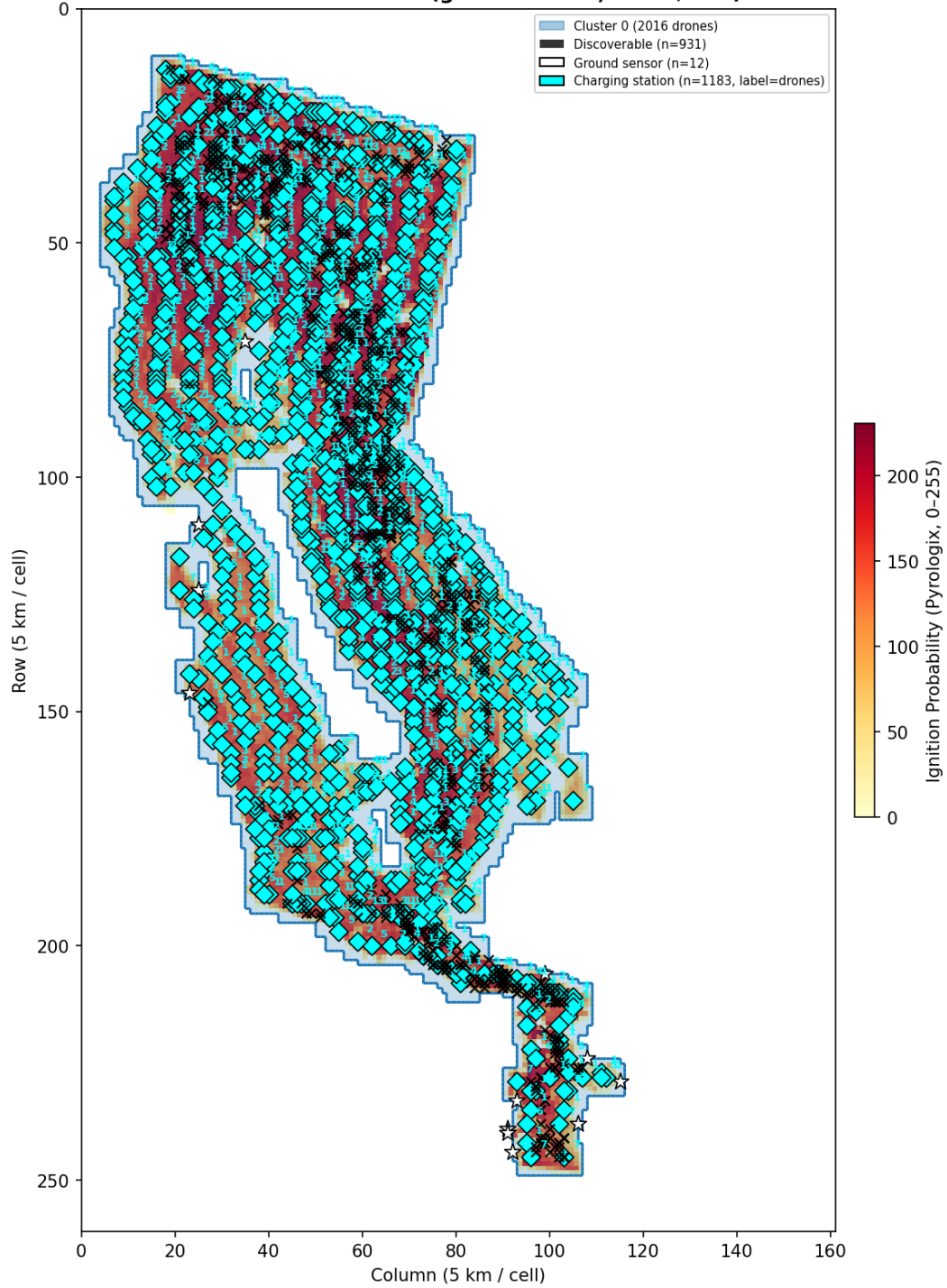


Figure 6: 500M placement — operational scale.

8. Detection Results

Routing benchmarks (TOP and MaxCoverage strategies) are pending for all three budget levels. With the corrected ILP, station spacing is now ≥ 5 cells (at \$100M), producing well-separated clusters with exactly 7 drones per station. This should yield efficient PSO routing with no bottleneck from oversized clusters.

9. How to Reproduce

Sensor placement only:

```
python-jl run_benchmark_california2021_yearly.py --sensor-only --budget 20
python-jl run_benchmark_california2021_yearly.py --sensor-only --budget 100
python-jl run_benchmark_california2021_yearly.py --sensor-only --budget 500
```

Generate placement plots:

```
python visualize_sensor_placement_2021.py \
    California2021Dataset/logs/sensor_alloc_GaussianBudget20M_TOP_261x161_mean.json \
    --scale both
python visualize_sensor_placement_2021.py \
    California2021Dataset/logs/sensor_alloc_GaussianBudget100M_TOP_261x161_mean.json \
    --scale both --tag _100M
python visualize_sensor_placement_2021.py \
    California2021Dataset/logs/sensor_alloc_GaussianBudget500M_TOP_261x161_mean.json \
    --scale both --tag _500M
```

Full benchmark (placement + routing):

```
python-jl run_benchmark_california2021_yearly.py --budget 20
python-jl run_benchmark_california2021_yearly.py --budget 100
python-jl run_benchmark_california2021_yearly.py --budget 500
```